Module Description

IN8005: Introduction into Computer Science (for non informatics studies)

Department of Computer Science

Module level:	Language:	Module duration:	Occurrence:
Bachelor	English	one semester	winter semester
Credits*:	**Total number of hours:**	**Self-study hours:**	**Contact hours:**
5	150	90	60

* The number of credits can vary depending on the corresponding SPO version. The valid number is always indicated on the Transcript of Records or the Performance Record.

Description of achievement and assessment methods:
Type of Assessment: written exam (90 minutes)
The exam takes the form of written test. Knowledge questions allow to assess acquaintance with and understanding of the basic concepts of Computer Science. Small programming and modelling problems allow to assess the ability to practically apply the learned programming- and query-languages and modelling-techniques for the solution of small problems.
Homework will be scored and upon achieving a minimum equired number of points, a 0,3 bonus for the final grade is granted.

In case of epidemiologic emergencies, the exam may be substituted by a graded electronic exercise or a proctered exam.

Possibility of re-taking:
In the next semester: No
At the end of the semester: Yes

(Recommended) requirements:
Recommended requirements are Mathematics modules of the first year of the TUM-BWL bachelor's program as well as the module WI000275 'Management Science'.

Contents:
The module IN8005 is concerned with topics such as:
- Database Management Systems,.ER models, Relational Algebra, SQL
- Java as a programming language:
++ basic constructs of imperative programming (if, while, for, arrays etc.)
++ object-oriented programming (inheritance, interfaces, polymorphism etc.)
++ basics of Exception Handling and Generics
++ code conventions
++ Java class library

- Basic algorithms and data structures:
++ algorithm concept, complexity
++ data structures for sequences (arrays, doubly linked lists, stacks & queues)
++ recursion
++ hashing (chaining, probing)
++ searching (binary search, balanced search trees)
++ sorting (Insertion-Sort, Selection-Sort, Merge-Sort)

Study goals:
Upon successful completion of the module, participants understand important foundations, concepts and ways of thinking of Computer Science, in particular object-oriented programming, databases and SQL, and basic algorithms and data structures, have an overview over these topics and be able use them for the development of own programs with a link to a database in a basic way.

Teaching and learning methods:
Lecture and practical tutorial assignments. A central tutorial deepens the understanding of the concepts introduced in the lecture using example assignments in regard to being able to solve given problems. In the tutorials, the students solve basic assignments under intensive supervision, which contributes to providing them with the basic skills in programming, in order to be able to apply the knowledge acquired by self-study of the accompanying materials of lecture and central tutorial for autonomously solving the programming assignments of the homework. During the second half of the semester, the students work on a small practical project, which aims at deepening the connected understanding of the desired learning outcomes. Programming aspects of this project are distributed over tutorial and homework assignments and are aligned with the topics of the respective week.

Media formats:
Slides, blackboard, lecture- and central tutorial recording, discussion boards in suitable e-learning platforms

Literature:
Chapters from textbooks, which are closely associated with the module content and are provided to the students online.

Responsible for the module:
Groh, Georg; Prof. Dr. rer. nat. habil.: grohg@mytum.de

Courses (Type, SH) Lecturer:

0000002036 Exercise Session for Introduction into Computer Science (for non Informatics studies, TUM BWL) (IN8005) (2SWS UE, WS 2024/25)
Groh G [L], Dall'Olio G, Groh G, Steinberger C

240905438 Introduction into Computer Science (for non Informatics studies, TUM BWL) (IN8005) (2SWS VO, WS 2024/25)
Groh G

For further information about this module and its allocation to the curriculum see:
https://campus.tum.de/tumonline/wbModHb.wbShowMHBReadOnly?pKnotenNr=458528

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